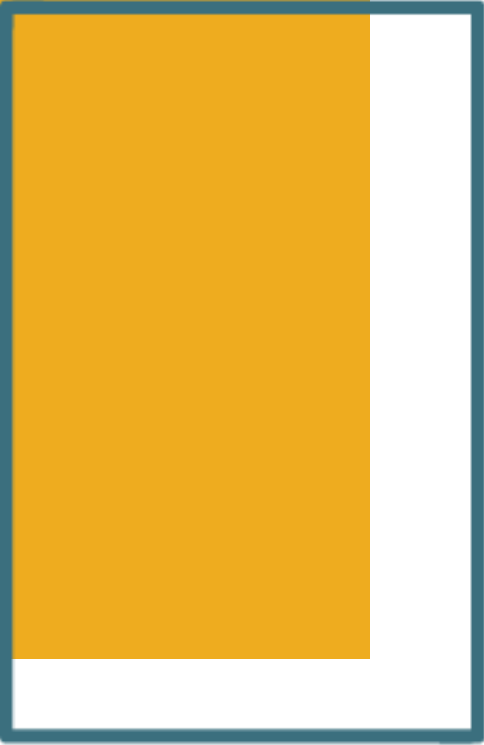




05

Video Basic with OpenCV



What is a digital video?

Video is an electric medium for the recording, copying, playback, broadcasting, and display of moving visual media. Video systems vary in display resolution, aspect ratio, refresh rate, color capabilities and other qualities. There are 2 types format of video; analog and digital. Digital video is an electronic representation of moving visual images (video) in the form of encoded digital data.

Characteristics of video

- Number of frames per second (fps) the numbers of still pictures per unit of time of video.
- Aspect ratio describe the proportional relationship the width and height of video screens.
- Refresh rate is how many times per second the display is able to draw new image, measured in Hertz (Hz).
- Color capabilities representation and maps encoded color values to visible colors reproduced by the system.

How to connect to camera?

- Use `cv2.VideoCapture()` to get a video capture object for the camera.
- Set up an infinite while loop and use the `read()` method to read the frames using the above created object.
- Use `cv2.imshow()` method to show the frames in the video.
- Breaks the loop when the user clicks a specific key.

```
import cv2

cap = cv2.VideoCapture(0)

width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))

while True:

    # Capture frame-by-frame
    ret, frame = cap.read()

    # Our operations on the frame come here
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # Display the resulting frame
    cv2.imshow('frame', gray)

    # This command let's us quit with the "q" button on a keyboard.
    # Simply pressing X on the window won't work!
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

# When everything done, release the capture and destroy the windows
cap.release()
cv2.destroyAllWindows()
```

How to using video file?

To capture a video, we need to create a `VideoCapture` object. `VideoCapture` have the device index or the name of a video file. Device index is just the number to specify which camera. If we pass 0 then it is for first camera, 1 for second camera so on. We capture the video frame by frame.

```
cv2.VideoCapture(0): Means first camera or webcam.
cv2.VideoCapture(1): Means second camera or webcam.
cv2.VideoCapture("file name.mp4"): Means video file
```

```
# Importing libraries
import cv2
import numpy as np

# Create a VideoCapture object and read from input file
cap = cv2.VideoCapture('tree.mp4')

# Check if camera opened successfully
if (cap.isOpened() == False):
    print("Error opening video file")

# Read until video is completed
while(cap.isOpened()):

    # Capture frame-by-frame
    ret, frame = cap.read()
    if ret == True:

        # Display the resulting frame
        cv2.imshow('Frame', frame)

        # Press Q on keyboard to exit
        if cv2.waitKey(25) & 0xFF == ord('q'):
            break

    # Break the loop
    else:
        break

# When everything done, release
# the video capture object
cap.release()

# Closes all the frames
cv2.destroyAllWindows()
```

Picture Source

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